

Suad Mohamed

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EDUCATION

Belmont University Nashville, TN

Bachelor of Science in Computer Science | Bachelor of Science in Psychology

GPA: 3.91, College of Science & Mathematics Dean's Research Award

HONORS AND AWARDS

- 2025 College of Science & Mathematics Dean's Research Award
- Belmont University Deans List 2021-25

RELEVANT COURSEWORK

Research Methods, Data Science of Behavior, Human Computer Interaction I, Intro to Artificial Intelligence, Cognitive Psychology, Learning & Conditioning, Psychological Testing

RESEARCH EXPERIENCE

Discerning Code Comprehension Quality From Programmer Visual Attention Feb 2024- Present

Dr. Yu Huang: Mind Lab | Vanderbilt University

- Analyzed 21+ hours of eye-tracking data from 27 CS students writing Java code summaries, incorporating fixation count, duration, scanpath and attention switching analysis
- Developed Transformer-based classification model integrating custom semantic category/duration embedding and CodeBert tokenization to predict comprehension quality from scanpath, demonstrating 8.1% improvement compared to baseline model without visual attention features.
- Conducted in-depth analysis including non-parametric pair-wise testing, multivariate analysis, and LMER to investigate visual attention indicators of code summary quality.

Beyond the Prompt: How Novices Engage with AI During Code Summarization May 2024-2025

Dr. Esteban Rodriguez & Dr. Michael Oliver | Belmont University

- Led design of within-subject eye tracking study examining how students (22) summarize code with and without LLM assistance, delegating recruitment protocol and analysis tasks across 4 undergraduates.
- Submitted full IRB protocol including recruitment, consent, eligibility screening, and risk procedures as PI of the study.
- Constructed experimental dataset filtering 150K Python methods and configured experimental materials including AOIs and split-screen task interface on Google's Gemini
- Designed analysis pipeline independently with faculty supervision, including repeated measures ANOVAs, hybrid summary evaluation, and Gemini prompt qualitative evaluation.

Chatting with AI: Deciphering Developer Conversations with ChatGPT Nov 2023-Apr 2024

Dr. Esteban Rodriguez & Dr. Michael Oliver | Belmont University

- Conducted qualitative analysis pipeline for 1,900+ developer conversations with ChatGPT, including open-coding scheme and inter-rater reliability statistics with final of 19 conversation categories.
- Designed four classical ML models (Support Vector Machine, Naïve Bayes, Random Forest, and Decision Tree) to explore automatic classification of developer-AI conversations.

Cognitive Processes in Programming: EEG Study on Memory & Attention Jan 2024-May 2024

Dr. Michael Oliver | Belmont University

- Designed and ran a multi-task EEG experiment combining working memory, attention, and linguistic processing task to explore cognitive processes in programming students.
- Collected and processed EEG and behavioral data across 25 undergraduate students, performed pairwise statistical tests by experience, and presented poster at Belmont University's Research Symposium.

RELEVANT WORK EXPERIENCE

Software Engineer I

Microsoft | Atlanta, GA

Aug 2025 - Present

- Design internal networking utilities for cloud service management and on call engineer improvements

Software Engineer Intern

Microsoft | Redmond, WA

May 2024 – Aug 2024

- Implemented microservice to address monitoring gap with Azure Container's CI testing.

Software Development Intern

IDEMIA | Brentwood, TN

May 2023 – Dec 2023

- Automated digital license verification and created integration tests to ensure feature reliability.

LEADERSHIP EXPERIENCE – Optional section

Chair of Research and Professional Development Student Advisory Board

Mar 2023-May 2025

Belmont University Department of CS and Mathematics | Nashville, TN

- Led initiatives to expand research opportunities by conducting faculty and student interviews and presenting actionable suggestions at departmental meetings.

Founder, Director of Programming

Jan 2023-May 2025

Bruins in Tech, Belmont University | Nashville, TN

- Organized workshop and panel discussions to facilitate mentorship opportunities to CS undergrads.

PUBLICATIONS – Optional section

Conference Papers

1. **Mohamed, S.**, Parvin, A., & Parra, E. (2024, April). Chatting with ai: Deciphering developer conversations with chatgpt. In *Proceedings of the 21st International Conference on Mining Software Repositories* (pp. 187-191).
2. **Mohamed, S.**, Ismail, N., Amaya Hernandez, K., Parvin, A., Oliver, M., & Parra, E. (2025, May). Design of An Eye-Tracking Study Towards Assessing the Impact of Generative AI Use on Code Summarization. In *Proceedings of the 2025 Symposium on Eye Tracking Research and Applications* (pp. 1-8).

Accepted:

3. **Mohamed, S.**, Parvin, A., Ismail, N., Amaya, K., Oliver, M., & Parra, E.. Beyond the prompt: How novices engage with AI during code summarization. In *Proceedings of International Conference on Software Engineering Software Engineering Education Track 2026*.

SKILLS OR LABORATORY SKILLS

Programming: Java, Python, R, C#, CSS, JavaScript, Razor, Docker, Kubernetes

Lab: Human Research, Rodent Animal Research